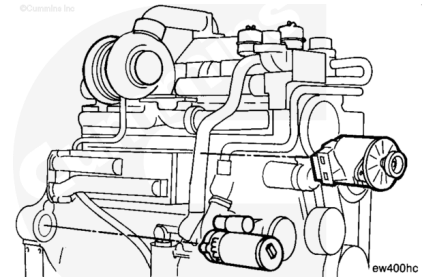


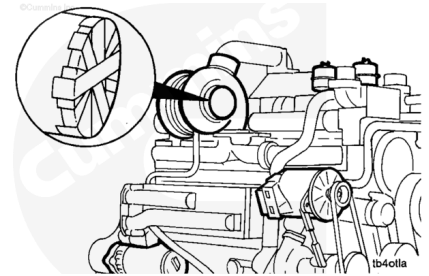
Trainings ▾

Preparatory Steps

Install all accessories and brackets that were removed from the engine.



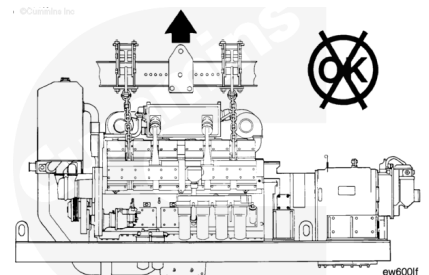
Uncover all the engine openings.



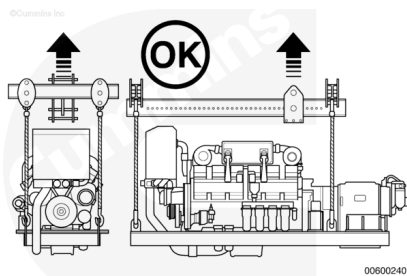
Install

⚠ CAUTION ⚠

Never use the engine lifting brackets or the engine lifting fixture, Part Number 3163264, to lift an engine when drive units or a radiator is mounted on common base rails. Never use the lifting bracket on the block to lift an engine that is mounted on base rails when there is a drive or radiator also mounted on the base rails. The lifting brackets or parts can break, causing the engine to fall. Engine damage can result.

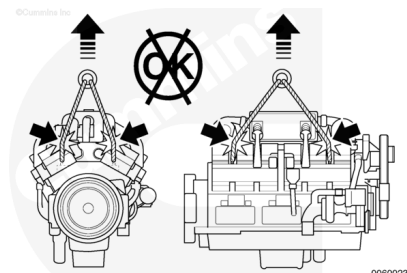


Always use the lifting brackets that are connected to the base rails when drive units or a radiator is mounted on the base rails with the engine.



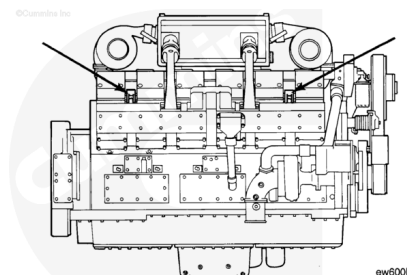
⚠ CAUTION ⚠

Do not attempt to lift the engine with an apparatus that allows side-loading of the lifting brackets, or allows the chain or cable to apply force to other parts of the engine. The lifting brackets or parts can break, causing the engine to fall and be damaged.



⚠ CAUTION ⚠

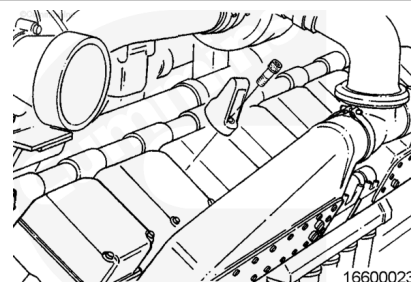
Before lifting the engine, check the four engine lifting brackets. Be sure each bracket contains a spot-face under the hardened flat washer. If the bracket is not spot-faced, remove the capscrew. Check for rust and corrosion. If the capscrew has rust or corrosion, replace the capscrew before lifting the engine.



KTA38GC, K38 and QSK38 engines have four lifting brackets, two on each bank. The lifting brackets are installed between cylinder numbers 1 and 2, and 5 and 6.

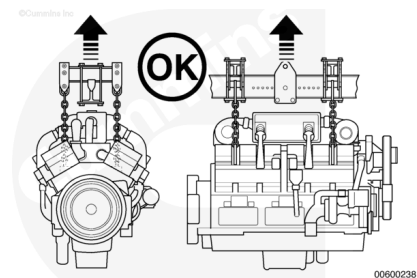
The lifting brackets for the K50 are installed between cylinder numbers 2 and 3, and 6 and 7.

Remove the lifting brackets. Check for cracks, rust and corrosion. Install lifting brackets. Refer to Procedure 016-001 (Engine Lifting Brackets) in Section 16.
(/qs3/pubsys2/xml/en/procedures/28/28-016-001-tr-gc.html)



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



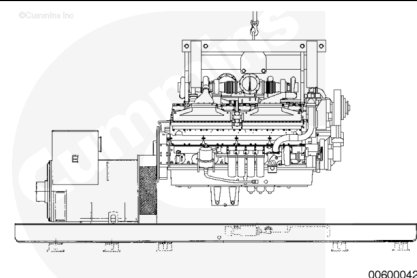
The engine lifting fixtures, Part Number 3163264, are designed to lift the engine **only**. Any additional components attached (including to the drive end, non-drive end and/or chassis) **must** be removed.

The lifting force **must** be in a straight vertical line with the lifting brackets in both directions.

Engine lifting fixture, Part Number 3163264, is available from Cummins Inc. This fixture is available to lift the KTA38GC, K38, K50, QSK38, QSK50 assembled engines.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Use a properly rated hoist and Part Number 3163264 Engine Lifting Fixture, to install the engine.

With the engine installed, check the crankshaft end clearance to make sure it is still within specification. Refer to Procedure 001-007 in Section 1.

(/qs3/pubsys2/xml/en/procedures/28/28-001-007-tr.html)

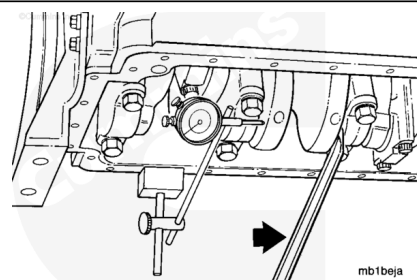
Remove an accessible hand hole cover. Refer to Procedure 001-035 in Section 1.

(/qs3/pubsys2/xml/en/procedures/28/28-001-035-tr.html)

The crankshaft can be moved forward and backward by using a pry bar on the crankshaft webs through the hand hole cover opening. The end clearance can be measured through the hand hole cover opening on a crankshaft counterweight or on the front damper.

Note : Do not use a pry bar on the damper to move the crankshaft. Doing so may cause damage to the damper.

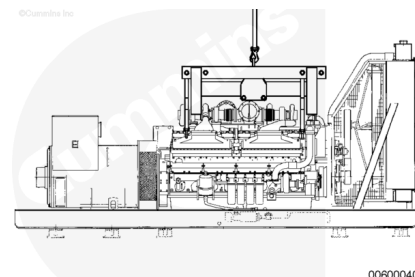
Note : The oil pan is shown removed from the engine for illustration purposes.



Once the end clearance is confirmed to be within specification, replace the hand hole cover. Refer to Procedure 001-035 in Section 1. (/qs3/pubsys2/xml/en/procedures/28/28-001-035-tr.html) If the end clearance is out of specification, check for misalignment with the attached drive units.

⚠ WARNING ⚠

Inspect the engine lifting brackets for damage or cracks. Do not attempt to lift the engine if any cracks or damage are visible. Using damaged lifting equipment can cause serious personal injury and/or property damage.



⚠ CAUTION ⚠

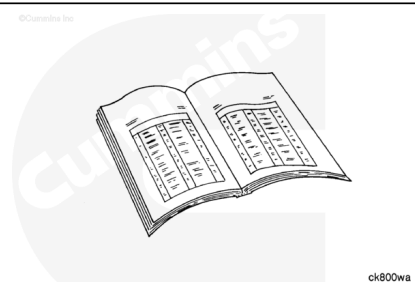
Make sure all lines, hoses, and tubes are in good condition, properly routed, and fastened to prevent engine damage.

Align the engine in the chassis on base rails, and tighten the engine mounting capscrews. See the OEM torque specifications.

Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



⚠ WARNING ⚠

Coolant is toxic. Keep away from small children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Fill the engine with clean lubricating oil. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/28/28-007-037-tr.html)
- Fill the cooling system with coolant. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html)
- Connect the air supply to the air starter, if equipped. Refer to Procedure 012-022 in Section 12.
(/qs3/pubsys2/xml/en/procedures/102/102-012-022.html)
- Install air conditioning lines, if needed. Refer to the OEM service manual.
- Connect the intake and exhaust system pipes. Refer to the OEM service manual.
- Install all chassis components necessary to install the engine to the equipment. Refer to the OEM service manual.

- Connect the starter cable, engine ground straps, cab, or chassis to engine hoses, tubing, electrical wires, and hydraulic lines.
- Connect the battery cables. Refer to the OEM service manual.
- Make final inspection to make sure all hoses, wires, linkages, and components have been properly installed and tightened.
- Operate engine at low idle for 2 to 3 minutes.
- Stop the engine and wait 5 to 7 minutes for the oil to drain to the oil pan. Check the oil and coolant levels again.
- Fill the engine to the correct oil and coolant levels, if necessary. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/28/28-007-037-tr.html>) Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html>)
- Operate the engine for 8 to 10 minutes to check for proper operation, unusual noises, and coolant, fuel, and lubricating oil leaks. Refer to Procedure 014-006 in Section 14. (</qs3/pubsys2/xml/en/procedures/28/28-014-006.html>)